

01

```
import java.util.*;

class Example{

    public static void main(String args[]){

        Scanner input = new Scanner(System.in);


        System.out.print("Enter first number: ");

        int num1 = input.nextInt();


        System.out.print("Enter second number: ");

        int num2 = input.nextInt();


        if(num1>num2){

            System.out.println("first num " + "+" + "second num " + "=" + (num1 +
num2));

        }else{

            System.out.println("num1 =" + num1);

            System.out.println("num2 =" + num2 );

        }

    }

}
```

02

```
import java.util.*;

class Example{

    public static void main(String args[]){
```

```
Scanner input = new Scanner(System.in);
```

```
System.out.print("Enter first number: ");
```

```
int num1 = input.nextInt();
```

```
if(num1<0){
```

```
    System.out.println("Absolute value = "+ -num1);
```

```
    }else{
```

```
        System.out.println("Absolute value = "+ num1);
```

```
    }
```

```
}
```

```
}
```

03

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Chemistry marks: ");
```

```
        int chemistry = input.nextInt();
```

```
        System.out.print("Physics marks: ");
```

```
        int physics = input.nextInt();
```

```

        System.out.print("Combined maths marks: ");

        int maths = input.nextInt();

        int total = chemistry + physics + maths;

        int avg = total/3;

        System.out.println("Total = " + total);
        System.out.println("Average = " + avg);

        if(avg>75){
            System.out.println("\nPass");
        }else{
            System.out.println("\nFail");
        }
    }
}

```

}

}

04

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Enter the unit price: ");
```

```
        double price = input.nextDouble();
```

```
System.out.print("Enter amount: ");

double amount = input.nextInt();

double total = price * amount;

System.out.println("Total = " + total);

if(total>1500){
    System.out.println("You are entitled to the super draw");
}else{
    System.out.println("Try again");
}
```

```
}
```

```
}
```

05

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Enter the unit price: ");
```

```
        double price = input.nextDouble();
```

```
        System.out.print("Enter amount: ");
```

```
        double amount = input.nextDouble();
```

```
double total = price * amount;
```

```
System.out.println("Total = " + total);
```

```
if(total>500){
```

```
    double discount = total * 0.05;
```

```
    double newtotal = total - discount;
```

```
    System.out.println("Discount = "+ discount);
```

```
    System.out.println("New total = "+ newtotal);
```

```
}else{
```

```
    System.out.println("No discount given");
```

```
}
```

```
}
```

```
}
```

06

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Enter the year: ");
```

```
        int year = input.nextInt();
```

```
        if ((year % 4 == 0 && year % 100 !=0) || (year % 400 == 0)){
```

```

        System.out.println("leap year");
    }
    else{
        System.out.println("Not a leap year");
    }
}
}

```

07

```

import java.util.*;

class Example{

    public static void main(String args[]){

        Scanner input = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int num1 = input.nextInt();

        System.out.print("Enter second number: ");
        int num2 = input.nextInt();

        System.out.print("Enter third number: ");
        int num3 = input.nextInt();

        int max = num1;

        if (num2 > max){
            max = num2;

```

```

    }
    if (num3 > max){
        max = num3;
    }
    System.out.println("Maximum number is: " + max);
}
}

```

08

- C. if(x==10){}
- E. if((x=100)!=10){}

09

- A. if(b){}
- C. if(b==false){}
- E. if((b=false)==false){}
- F. if(b=(false==true)){}

10

```

import java.util.*;

class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);

        System.out.print("Enter number: ");

        int num = input.nextInt();

        if (num % 2 == 0){

```

```

        System.out.println("Your number is odd");
    }
    else{
        System.out.println("Your number is even" );
    }
}
}

```

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- A. System.out.println(b); // true
- B. System.out.println(b=true); // true
- C. System.out.println(b==true); //true
- D. System.out.println(b!=true); // false
- E. System.out.println(b=true==true); // true
- F. System.out.println((b=true)==false); // false
- G. System.out.println(b=(true!=false)); // true

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- A. System.out.println(x=9); // 9
- B. System.out.println(x==9); // false
- C. System.out.println(x=9!=10); // error
- D. System.out.println((x=9)==10); // false
- E. System.out.println((x=9)<=10); // true

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```
int x=20,y=60;
```

```
boolean bool;
```

```
System.out.println(x=10); // 10
```

```
System.out.println(bool=true); // true
```



```
System.out.println(x=10>0); // error
System.out.println((x=10)>0); // true
System.out.println(bool=(x=10)>0); // true
System.out.println(bool=x+y>100); // false
```

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```
byte b = 10;
short s = 100;
int x = 125;
long l = 15000;
float f = 1.5f;
double d = 21.231;
char c = 'c';
boolean bool = 10>9;
System.out.println(b+s+x+""+f+d+c+bool); // 2351.521.231ctrue
System.out.println(""+b+s+x+f+d+c+bool); // 101001251.521.231ctrue
System.out.println(b+s+x+f+d+c+""+bool); // 356.731true
System.out.println(b+s+x+f+d+c+bool+""); // compile error
System.out.println(bool+b+f+d+c+""+x+l); // compile error
```

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```
System.out.println( i ==1); // true
System.out.println( j ==3); // false
System.out.println( ( i >=1) && ( j <4) ); // true
System.out.println( ( m <=99) & ( k < m ) ); // false
System.out.println( ( j >= i ) || ( k == m ) ); // true
System.out.println( ( k + m < j )||(3- j>= k)); // false
System.out.println( !( k > m ) ); // false
```

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A. `int x=1; // 1 2 3`

B. `int x=2; // 2 3`

C. `int x=3; // 3`

D. `int x=4; // 4 1 2 3`

E. `int x=0; // 4 1 2 3`

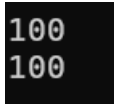
F. `int x=5; // 4 1 2 3`

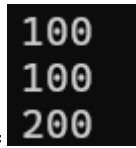
17

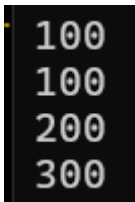
D. Prints 0 0

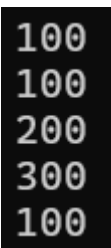
18

A. Line 1 = 100 legal

B. Line 2 =  legal

C. Line 3 =  legal

D. Line 4  legal

E. Line 5  legal

```
100
100
200
300
100
200
```

F. Line 6 legal

G. Line 7 = error, z is out of scope

```
100
100
200
300
100
200
100
```

H. Line 8 legal

I. Line 9 =error, out of scope

K. Line 10 = error, out of scope

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A. int x=1; // 1

B. int x=2; // 2 3 1

C. int x=3; // 3 1

D. int x=4; // Wrong

E. int x=0; // Wrong

F. int x=5; // Wrong

20

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        Scanner input = new Scanner(System.in);
```

```

        System.out.print("Enter number: ");
        int num = input.nextInt();

        if (num < 0){
            System.out.println("Negative integer");
        }
        else if(num > 0){
            System.out.println("Positive integer");
        }else{
            System.out.println("Zero");
        }
    }
}

```

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```

import java.util.*;

class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int num1 = input.nextInt();

        System.out.print("Enter second number: ");
        int num2 = input.nextInt();
    }
}

```

```

int diff;

if (num1 > num2){
    diff = num1 - num2;
}else{
    diff = num2 - num1;
}

    System.out.println("Absolute difference is: "+ diff);
}
}

```

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- A. char x='A';
- B. int x=65;
- C. int x=65536;
- D. byte x=65;
- E. short x=66 ;

23

```

import java.util.*;

class Example{

    public static void main(String args[]){

        Scanner input = new Scanner(System.in);

        System.out.print("Enter first number: ");

        int num1 = input.nextInt();

        System.out.print("Enter second number: ");
    }
}

```

```

        int num2 = input.nextInt();

        if (num1 < num2){
            System.out.println("The first number is less than the second
number");
        }else if(num1 > num2){
            System.out.println("The first number is greater than the second
number");
        }else{
            System.out.println("Both are equal");
        }
    }
}

```

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```

import java.util.*;

class Example{

    public static void main(String args[]){

        Scanner input = new Scanner(System.in);

        System.out.print("Enter first number: ");

        int num1 = input.nextInt();

        System.out.print("Enter second number: ");

        int num2 = input.nextInt();

        System.out.print("Enter third number: ");

        int num3 = input.nextInt();
    }
}

```

```
int rmd1 = num1 % 10;
```

```
int rmd2 = num2 % 10;
```

```
int rmd3 = num3 % 10;
```

```
if(rmd1 == rmd2 || rmd1 == rmd3 || rmd2 == rmd3){
```

```
    System.out.println("true");
```

```
    }
```

```
}
```

```
}
```

25

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Enter first number: ");
```

```
        int num1 = input.nextInt();
```

```
        System.out.print("Enter second number: ");
```

```
        int num2 = input.nextInt();
```

```
        System.out.print("Enter third number: ");
```

```
        int num3 = input.nextInt();
```

```
        if(num1 > (num2 - num3) || num1 > (num3 - num2)){
```

```

        System.out.println("True");
    }else if(num2 > (num1 - num3) || num2 > (num3 - num1)){
        System.out.println("True");
    }else if(num3 > (num1 - num2) || num3 > (num2 - num1)){
        System.out.println("True");
    }else{
        System.out.println("False");
    }
}
}

```

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```

import java.util.*;

class Example{

    public static void main(String args[]){

        Scanner input = new Scanner(System.in);

        System.out.print("Selling price: ");
        double sell_price = input.nextDouble();

        System.out.print("Cost of the product: ");
        double cop = input.nextDouble();

        double profit = sell_price - cop;
        double loss = cop - sell_price;

        if (sell_price > cop){

```



```

        System.out.println("Profit = " + profit);
    }else if (cop > sell_price){
        System.out.println("Loss = " + loss);
    }else{
        System.out.println("No Profit No Loss");
    }
}
}

```

27

```

import java.util.*;

class Example{

    public static void main(String args[]){

        Scanner input = new Scanner(System.in);

        System.out.print("Input first number: ");
        int num1 = input.nextInt();

        System.out.print("Input second number: ");
        int num2 = input.nextInt();

        System.out.print("Input third number: ");
        int num3 = input.nextInt();

        if (num1 < num2 && num2 < num3){

```

```

        System.out.println("Increasing");
    }else if (num1 > num2 && num2 > num3){
        System.out.println("Decreasing");
    }else{
        System.out.println("Neither increasing nor decreasing order");
    }
}
}

```

28

```

import java.util.*;

class Example{

    public static void main(String args[]){

        Scanner input = new Scanner(System.in);

        System.out.print("Enter age: ");

        int age = input.nextInt();

        System.out.print("Enter weight (kg): ");

        int weight = input.nextInt();

        if(age >= 18 && weight >= 50){

            System.out.println("Eligible for blood donation");

        }else{

            System.out.println("Not eligible for blood donation");

        }

    }
}

```

```
}
```

29

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Enter first number: ");
```

```
        int num1 = input.nextInt();
```

```
        System.out.print("Enter second number: ");
```

```
        int num2 = input.nextInt();
```

```
        if((num1 < 0 && num2 < 0) || (num1 > 0 && num2 > 0)){
```

```
            System.out.println("True");
```

```
        }else{
```

```
            System.out.println("False");
```

```
        }
```

```
    }
```

```
}
```

30

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        Scanner input = new Scanner(System.in);
```

```

        System.out.print("Enter a character: ");

        char ch = input.next().charAt(0);

        if (ch >= 'a' && ch <= 'z') {
            System.out.println("Lowercase");
        } else if (ch >= 'A' && ch <= 'Z') {
            System.out.println("Uppercase");
        } else {
            System.out.println("Not an alphabet character");
        }
    }
}

33
import java.util.*;

class Example{

    public static void main(String args[]){

        Scanner input = new Scanner(System.in);

        System.out.print("Enter salary: ");

        double salary = input.nextDouble();

        System.out.print("Enter year of service: ");

        double service = input.nextDouble();

        if(service < 5){

            System.out.println("Total salary = " + salary * 1.1);

```

```

        }else if(service < 10){
            System.out.println("Total salary = " + salary * 1.15);
        }else{
            System.out.println("Total salary = " + salary * 1.25);
        }
    }
}

```

38

```

import java.util.*;

class Example{

    public static void main(String args[]){

        Scanner input = new Scanner(System.in);

        System.out.print("Enter attendance rate: ");

        int attendance = input.nextInt();

        System.out.print("Enter average marks: ");

        int avg = input.nextInt();

        if((attendance > 80) && (avg > 50)){

            System.out.println("Eligible to sit for the O/L examination");

        }else{

            System.out.println("Not eligible to sit for the O/L examination");

        }

    }

}

```

39

```
import java.util.*;

class Example{

    public static void main(String args[]){

        Scanner input = new Scanner(System.in);

        System.out.print("Enter time (00.00-24.00): ");

        double time = input.nextDouble();

        if(time >= 0 && time < 12){

            System.out.println("Good morning");

        }else if(time >=12 && time < 16){

            System.out.println("Good afternoon");

        }else if(time >=16 && time < 19){

            System.out.println("Good evening");

        }else if(time >=19 && time <= 24){

            System.out.println("Good night");

        }else{

            System.out.println("Invalid time");

        }

    }

}
```

42

```
import java.util.*;

class Example{

    public static void main(String args[]){

        Scanner input = new Scanner(System.in);

        System.out.print("Enter waist size(inches-UK): ");

        double size = input.nextDouble();

        if(size >=28 && size <=29){

            System.out.println("XS");

        }else if(size >= 30 && size <= 31){

            System.out.println("S");

        }else if(size >= 32 && size <= 34){

            System.out.println("M");

        }else if(size >= 36 && size <= 38){

            System.out.println("L");

        }else if(size >= 40 && size <= 42){

            System.out.println("XL");

        }else{

            System.out.println("Invalid size");

        }

    }

}
```

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```
import java.util.*;

class Example{

    public static void main(String args[]){

        Scanner input = new Scanner(System.in);

        System.out.print("Number of day: ");

        int day = input.nextInt();

        switch(day){

            case 6:

            case 13:

            case 20:

            case 27:

                System.out.println("Sunday");

            break;

            case 7:

            case 14:

            case 21:

            case 28:

                System.out.println("Monday");

            break;

            case 1:

            case 8:

            case 15:

            case 22:
```



```
        System.out.println("Tuesday");
    break;
    case 2:
    case 9:
    case 16:
    case 23:
        System.out.println("Wednesday");
    break;
    case 3:
    case 10:
    case 17:
    case 24:
        System.out.println("Thursday");
    break;
    case 4:
    case 11:
    case 18:
    case 25:
        System.out.println("Friday");
    break;
    case 5:
    case 12:
    case 19:
    case 26:
        System.out.println("Saturday");
    break;
```

default:

System.out.println("Invalid day");

}

}

}